Notes: Constantinides, Jackwerth, & Perrakis (RFS, 2009)
Mispricing of S&P 500 Index Options

Contents

1 Big picture

- **Question.** Are 1-month S&P 500 index call options rationally priced once the test allows for incomplete markets, transaction costs, bid-ask spreads, and nonparametric return distributions?
- **Core idea.** The paper does not ask whether Black-Scholes-Merton (BSM) fits the option surface. It asks whether there exists at least one risk-averse marginal trader whose pricing kernel is positive and decreasing in the market return.
- **Mispricing criterion.** A stochastic dominance violation means a trader can improve expected utility through a zero-net-cost trade in the index, bond, and options, net of transaction costs.
- **Data.** The sample uses monthly cross sections of 1-month S&P 500 index calls from 1986–2006. The 1986–1995 data come from the Berkeley Options Database; the 1997–2006 data come from OptionMetrics. There is no 1996 option sample.
- **Return distributions.** The authors estimate the physical distribution of 30-day index returns nonparametrically using historical and forward-looking samples, and also conditional distributions based on GARCH(1,1), ATM implied volatility, and VIX.
- **Main result.** Violations are widespread. Even with generous transaction costs, many monthly option cross sections cannot be rationalized by a monotone decreasing pricing kernel.
- **Precrash result.** Although pre-October-1987 option prices look reasonably BSM-like, they still violate stochastic dominance when the physical index-return distribution is estimated from time-series data.
- **Postcrash result.** Violations are not simply a "left-tail problem" or a volatility-smile-too-steep problem. OTM calls also generate substantial violations.
- **Time pattern.** Violations decline during 1988–1995 but rise again during 1997–2003. This casts doubt on the claim that the options market simply became more rational over time.
- **Economic takeaway.** Allowing for incomplete markets and realistic trading frictions does not fully rescue option prices. The evidence points to violations of broad economic restrictions, not merely failure of a parametric BSM formula.

2 Data and measurement

2.1 Why stochastic dominance is the right object

- Absence of arbitrage only requires a positive pricing kernel. Standard representative-agent asset pricing adds a stronger condition: for a risk-averse marginal investor whose wealth is increasing in the market, the pricing kernel should be decreasing in the market return.
- The paper calls a failure of this monotonicity-based restriction a **stochastic dominance violation**.
- The test is deliberately weaker than assuming a representative agent, complete markets, lognormal returns, or a specific utility function.

Interpretation. The test asks whether option prices can be supported by any plausible risk-averse marginal trader in a broad class. If the answer is no, the violation is economically stronger than simply rejecting BSM.

2.2 Option sample

- The paper studies **1-month European-style S&P 500 index calls**.
- It focuses on liquid calls with moneyness K/S between 0.90 and 1.05.
- For 1986–1995, the authors use the tick-by-tick Berkeley Options Database and select a monthly cross section with about 30 days to expiration.
- For 1997–2006, the authors use OptionMetrics and construct a hypothetical noon cross section from closing prices and the index observed at noon and close.
- The effective time periods are:
  - A: May 1986–October 1987, 15 months;
  - B: July 1988–March 1991, 29 months;

- C: April 1991–August 1993, 28 months;
- D: September 1993–December 1995, 26 months;
- E: February 1997–December 1999, 35 months;
- F: February 2000–May 2003, 37 months;
- G: June 2003–May 2006, 36 months.

Interpretation. The split matters because panels A–D use higher-quality Berkeley data, while panels E–G use OptionMetrics. The authors are cautious when comparing across these two data sources.

2.3 Index-return distributions

- The index return is a 30-calendar-day, roughly 21-trading-day, S&P 500 return.
- The unconditional distribution is estimated from four samples:
  - 1928–1986 historical returns, annualized volatility 21.0%;
  - 1972–1986 historical returns, annualized volatility 15.8%;
  - 1987–2006 forward-looking returns including the October 1987 crash, annualized volatility 15.2%;
  - 1988–2006 forward-looking returns excluding the crash, annualized volatility 14.8%.
- The paper also constructs conditional distributions using:
  - semiparametric GARCH(1,1);
  - the Black-Scholes implied volatility of the closest ATM 1-month option;
  - the revised VIX index, available from 1990 onward.
- The nonparametric state space has 61 values from $e^{-0.60}$ to $e^{0.60}$ in log spacing.

Interpretation. The paper intentionally avoids imposing lognormality. The disagreement between option prices and these physical distributions is the object of the test.

2.4 Transaction costs and bid-ask spreads

- Index trading faces a one-way proportional transaction cost of 50 basis points.
- Option trading costs are modeled in two ways:
  - **fixed costs**: 5, 10, or 20 basis points of the index price per call;
  - **proportional costs**: the ATM call cost is 5, 10, or 20 basis points of the index price, and other calls' costs scale with option price.
- Fixed costs tend to be high for cheap OTM calls and low for expensive ITM calls. Proportional costs tend to do the reverse.

Interpretation. Showing similar patterns under both cost schedules makes it harder to dismiss the results as an artifact of one assumed trading-cost schedule.

3 Models and empirical specifications

3.1 Economic environment

There is a stock/index with ex-dividend price S_t, dividend yield δ_t, and a risk-free bond paying gross return R. Traders can also trade European call options at date 0. Stock trading incurs proportional cost k:

$$\text{buy one share costs } (1+k)S_t, \qquad \text{sell one share yields } (1-k)S_t.$$

For option j, the trader buys at $P_j + k_j$ and sells at $P_j - k_j$. Thus $2k_j$ captures the bid-ask spread plus round-trip option trading costs.

Interpretation. The trader is not assumed to be the whole market. The test asks whether at least one trader in this class could be marginal in these securities.

3.2 Single-period stochastic dominance restrictions

Let S_{1i} be the end-of-period ex-dividend index level in state i with probability π_i, ordered so that S_{1i} is increasing in i. Let X_{ij} be option j's payoff in state i. Let $M^B(0), M^S(0)$ denote beginning-of-period marginal utilities from bond and stock accounts, and $M_i^B(1), M_i^S(1)$ the corresponding end-of-period marginal utilities. The paper tests feasibility of the following linear inequalities:

$$M^B(0) > 0, \qquad M^S(0) > 0, \tag{1}$$

$$M_i^B(1) > 0, \qquad i = 1, \ldots, I, \tag{2}$$

$$M_1^S(1) \geq M_2^S(1) \geq \cdots \geq M_I^S(1) > 0, \tag{3}$$

$$(1-k)M^B(0) \leq M^S(0) \leq (1+k)M^B(0), \tag{4}$$

$$(1-k)M_i^B(1) \leq M_i^S(1) \leq (1+k)M_i^B(1), \qquad i = 1, \ldots, I, \tag{5}$$

$$M^B(0) = R\sum_{i=1}^{I} \pi_i M_i^B(1), \tag{6}$$

$$M^S(0) = \sum_{i=1}^{I} \pi_i \left[\frac{S_{1i}}{S_0} M_i^S(1) + \frac{\delta S_{1i}}{S_0} M_i^B(1)\right], \tag{7}$$

$$(P_j - k_j)M^B(0) \leq \sum_{i=1}^{I} \pi_i M_i^B(1) X_{ij} \leq (P_j + k_j)M^B(0), \quad j = 1, \ldots, J. \tag{8}$$

Interpretation. The inequalities impose positivity, monotonicity, transaction-cost wedges, and pricing of the bond, stock, and calls. If the system is feasible, stochastic dominance violations are ruled out for that month. If it is infeasible, at least one zero-net-cost utility-improving trade exists.

3.3 Two-period version

The single-period model rules out trading during the 1-month life of the option. The two-period version allows one intermediate trading date, splitting the 30-day horizon into two 15-day periods.

Interpretation. Intermediate trading gives traders more ways to exploit mispricing. Empirically, allowing intermediate trading usually increases violations, so it strengthens rather than weakens the evidence.

3.4 Multiperiod option bounds

For a single call option, the paper also uses stochastic-dominance bounds that do not depend on the allowed trading frequency in the index and bond.

The upper bound is:

$$\bar{c}(S_t, t) = \frac{1+k}{(1-k)R_S^{T-t}} E\left([S_T - K]^+ \mid S_t\right),$$

where R_S is the expected return on the stock per period.

With the additional assumption that at least one trader's horizon coincides with option expiration, the lower bound is:

$$\underline{c}(S_t, t) = (1+\delta)^{t-T} S_t - \frac{K}{R^{T-t}} + \frac{E\left[(K - S_T)^+ \mid S_t\right]}{R_S^{T-t}}.$$

Interpretation. A bid price above the upper bound means the option is too expensive to write; an ask price below the lower bound means the option is too cheap to buy. These bounds allow the call market to be segmented because each option is tested one at a time.

4 Identification and empirical design

4.1 What identifies mispricing

- The paper identifies mispricing from the nonexistence of a decreasing marginal-utility/pricing-kernel vector that can support the observed index, bond, and option prices.
- It does not rely on BSM, constant volatility, or a parametric return distribution.
- It incorporates bid-ask spreads and transaction costs, so violations are not simply tiny arbitrage deviations.

Interpretation. The identifying comparison is between observed option prices and the broad set of prices that could be rationalized by risk-averse marginal traders under the estimated physical return distribution.

4.2 Why the design is conservative

- The test only requires one marginal trader in the modeled class; not all agents need to satisfy the assumptions.
- It allows heterogeneous agents and incomplete markets.
- It allows generous transaction costs.
- It uses several return-distribution estimates, including some based on option-implied volatility.
- The bound tests allow option-market segmentation by testing one strike at a time.

Interpretation. These choices make the rejection more meaningful. If violations survive under these allowances, they are not just a narrow rejection of a stylized complete-market model.

4.3 What the paper cannot fully prove

- The physical return distribution is estimated, not observed. The authors try many alternatives, but no finite set exhausts all possible beliefs.
- OptionMetrics data after 1997 may be noisier than Berkeley data. The paper partly addresses this by showing that many later violations are asymmetric, mostly upper-bound violations rather than random upper and lower mistakes.
- The modeled trader must have state-independent utility and end-of-period wealth increasing in the market return. The authors relax several related assumptions, but this class of traders is still the benchmark.

5 Tables and figures

5.1 Tables 1A and 1B: baseline feasibility under proportional and fixed option costs

Table 1A
Percentage of months without stochastic dominance violations with (A) proportional transaction costs and (B) fixed transaction costs

	A: 8605–8710	B: 8807–9103	C: 9104–9308	D: 9309–9512	E: 9702–9912	F: 0002–0305	G: 0306–0605
(A) Proportional transaction costs[a]							
Number of months	15	29	28	26	35	37	36
Unconditional index return distribution,	67	24	21	0	6	11	6
1928–1986	**73** (5)	**76** (10)	**50** (11)	**4** (5)	**26** (9)	**27**(6)	**17**(4)
	80	90	82	35	60	49	39
Unconditional index return distribution,	27	28	32	0	3	5	3
1972–1986	**53**	**48**	**54**	**15**	**9**	**8**	**11**
	67	76	93	81	29	19	47
Unconditional index return distribution,	13	38	43	0	0	5	3
1987–2006	**20**	**59**	**68**	**35**	**9**	**11**	**14**
	33	76	96	88	14	16	47
Unconditional index return distribution,	7	38	39	0	0	5	3
1988–2006	**20**	**55**	**71**	**27**	**0**	**8**	**11**
	47	66	93	81	20	16	61
Conditional index return distribution,	13	34	61	42	0	8	6
1972–2006, based on GARCH (1,1)	**27**	**59**	**82**	**69**	**11**	**27**	**11**
	33	76	86	96	29	43	47
Conditional index return distribution,	53	55	71	42	6	14	3
1972–2006, based on implied vol.	**87**	**83**	**96**	**73**	**29**	**38**	**19**
	100	93	86	96	71	84	50
Conditional index return distribution,	N/A	N/A	61	23	6	8	3
1972–2006, based on VIX			**86**	**54**	**17**	**24**	**19**
			89	92	51	86	58

[a]The panel displays the percentage of months in which stochastic dominance violations are absent in the cross section of option prices. The one-way transaction costs rate on the index is 50 bps. The transaction costs on the options are proportional. In each row, the one-way transaction costs rate on the ATM calls is 5 bps of the index price (top entries), 10 bps (bold, middle entries), or 20 bps (bottom entries). The numbers inside parentheses in the first row are bootstrap standard deviations of the first-row entries, based on 200 runs.

Table 1B
A percentage of months without stochastic dominance violations with (A) proportional transaction costs and (B) fixed transaction costs

	A: 8605–8710	B: 8807–9103	C: 9104–9308	D: 9309–9512	E: 9702–9912	F: 0002–0305	G: 0306–0605
(B) Fixed transaction costs[b]							
Number of months	15	29	28	26	35	37	36
Unconditional index return distribution,	47	7	0	0	9	3	0
1928–1986	**67**	**41**	**7**	**0**	**31**	**35**	**11**
	87	86	46	0	74	51	25
Unconditional index return distribution,	33	21	4	4	3	8	3
1972–1986	**53**	**52**	**32**	**4**	**11**	**16**	**14**
	60	72	68	15	29	24	53
Unconditional index return distribution,	13	38	21	0	6	11	8
1987–2006	**40**	**55**	**54**	**8**	**20**	**11**	**17**
	47	66	82	38	31	16	58
Unconditional index return distribution,	13	31	21	0	0	5	6
1988–2006	**33**	**55**	**43**	**4**	**11**	**11**	**17**
	47	69	71	35	17	11	53
Conditional index return distribution,	20	38	36	42	3	8	6
1972–2006, based on GARCH (1,1)	**40**	**52**	**61**	**85**	**17**	**24**	**19**
	40	79	89	100	34	38	58
Conditional index return distribution,	67	48	54	62	14	19	6
1972–2006, based on implied vol.	**100**	**79**	**89**	**88**	**37**	**38**	**28**
	100	93	96	96	74	78	72
Conditional index return distribution,	N/A	N/A	21	15	6	3	0
1972–2006, based on VIX			**75**	**54**	**17**	**19**	**28**
			93	96	46	73	72

[b]The panel displays the percentage of months in which stochastic dominance violations are absent in the cross section of option prices. The one-way transaction costs rate on the index is 50 bps. The transaction costs on the options are fixed. In each row, the one-way transaction costs rate on the calls is 5 bps of the index price (top entries), 10 bps (bold, middle entries), or 20 bps (bottom entries).

Read:

- The entries report the percentage of months *without* stochastic dominance violations. Low numbers mean more violations.
- With proportional option costs and the 1928–1986 return distribution, the precrash period A is feasible in 73% of months at the middle 10 bps option-cost assumption. Therefore, even the precrash BSM-looking period violates in about 27% of months.
- Postcrash panels often have much lower feasibility. For example, in Table 1A with the same row and middle cost assumption, panel D is feasible only 4%, and panels E–G are feasible only 26%, 27%, and 17%.
- The IV-based distribution helps, especially precrash and in some 1991–1995 periods, but it does not eliminate violations. Under proportional costs, the IV-based row is feasible only 29%, 38%, and 19% in panels E–G.
- Fixed option costs generate a similar pattern. The exact cells change, but widespread violations remain.

Economic message: The broad economic restriction fails often. Even a flexible physical distribution and generous trading costs cannot rationalize many option-price cross sections.

5.2 Tables 2A and 2B: ITM versus OTM calls

Table 2A

A percentage of months without stochastic dominance violations with (A) proportional transaction costs—ITM and OTM calls separately and (B) fixed transaction costs—ITM and OTM calls separately

	A: 8605–8710	B: 8807–9103	C: 9104–9308	D: 9309–9512	E: 9702–9912	F: 0002–0305	G: 0306–0605
(A) Proportional transaction costs—ITM and OTM calls separately[a]							
Number of months	15	29	28	26	35	37	36
Unconditional index return	87	90	79	38	69	46	42
distribution, 1928–1986	73	79	79	19	29	30	17
Unconditional index return	73	52	89	77	26	16	53
distribution, 1972–1986	53	45	75	46	11	14	11
Unconditional index return	53	66	89	88	20	16	50
distribution, 1987–2006	20	59	89	62	9	11	17
Unconditional index return	53	62	93	77	17	14	47
distribution, 1988–2006	20	59	89	54	3	8	14
Conditional index return distribution,	53	69	86	100	31	38	44
1972–2006, based on GARCH (1,1)	27	59	79	69	11	30	11
Conditional index return distribution,	100	97	93	100	80	81	50
1972–2006, based on implied vol.	87	83	96	77	34	59	19
Conditional index return distribution,	N/A	N/A	96	96	69	76	56
1972–2006, based on VIX			93	69	34	49	19

[a]The panel displays the percentage of months in which stochastic dominance violations are absent in the cross section of ITM calls (top entry) and OTM calls (bottom entry). The one-way transaction costs rate on the index is 50 bps. The one-way transaction costs rate on the index is 50 bps. The transaction costs on the options are proportional; for the ATM calls they are 10 bps of the index price.

Table 2B

A percentage of months without stochastic dominance violations with (A) proportional transaction costs—ITM and OTM calls separately and (B) fixed transaction costs—ITM and OTM calls separately

	A: 8605–8710	B: 8807–9103	C: 9104–9308	D: 9309–9512	E: 9702–9912	F: 0002–0305	G: 0306–0605
(B) Fixed transaction costs—ITM and OTM calls separately[b]							
Number of months	15	29	28	26	35	37	36
Unconditional index return	80	55	7	0	54	43	11
distribution, 1928–1986	80	90	86	27	57	38	39
Unconditional index return	60	59	39	4	14	16	14
distribution, 1972–1986	67	66	93	81	29	19	58
Unconditional index return	47	62	54	15	17	14	22
distribution, 1987–2006	47	62	93	96	23	16	44
Unconditional index return	40	59	50	15	17	11	22
distribution, 1988–2006	40	62	96	92	20	16	33
Conditional index return distribution,	40	59	64	88	29	30	28
1972–2006, based on GARCH (1,1)	47	62	82	96	31	41	53
Conditional index return distribution,	100	93	89	88	77	65	44
1972–2006, based on implied vol.	100	93	100	96	69	81	61
Conditional index return distribution,	N/A	N/A	79	77	26	38	33
1972–2006, based on VIX			93	96	60	76	64

[b]The panel displays the percentage of months in which stochastic dominance violations are absent in the cross section of ITM calls (top entry) and OTM calls (bottom entry). The one-way transaction costs rate on the index is 50 bps. The one-way transaction costs rate on the index is 50 bps. The transaction costs on the options are fixed as 10 bps of the index price.

Read:

- Table 2 separates in-the-money and out-of-the-money calls. Each cell has an ITM entry and an OTM entry.
- Under proportional costs, OTM calls often have lower feasibility than ITM calls, so OTM calls generate many violations.
- Under fixed costs, OTM transaction costs are effectively higher and OTM violations fall, while ITM violations rise. Still, OTM violations remain substantial.
- With the IV-based distribution, violations fall for both ITM and OTM calls, but do not disappear.

Economic message: The mispricing is not only about the left tail of the index-return distribution. Substantial OTM call violations contradict the simple interpretation that postcrash smiles are merely too steep because markets overprice crash insurance.

5.3 Tables 3 and 4: implied-volatility offsets and intermediate trading

Table 3
Percentage of months without stochastic dominance violations using conditional implied-volatility-based index return distributions with ±2% offset

	A: 8605–8710	B: 8807–9103	C: 9104–9308	D: 9309—9512	E: 9702–9912	F: 0002–0305	G: 0306–0605
Number of months	15	29	28	26	35	37	36
Implied vol. –2%	13	55	71	50	0	5	0
Implied vol. −1%	47	72	93	69	29	30	6
Implied vol.	87	83	96	73	29	38	19
Implied vol. +1%	93	76	96	65	26	32	19
Implied vol. +2%	100	72	86	65	23	30	19
Best of above	**100**	**83**	**96**	**73**	**29**	**38**	**19**

The table displays the percentage of months in which stochastic dominance violations are absent in the cross section of option prices. The one-way transaction costs rate on the index is 50 bps. The one-way transaction costs on the options are proportional. The one-way transaction costs rate on the ATM calls is 10 bps of the index price. All results use the conditional implied-volatility-based index return distribution over the 1972–2006 period. Four offsets are used to change the implied ATM volatility by −2, −1, 1, or 2% annualized return. The bold results "Best of above" count a monthly cross section as feasible if feasibility is established either without implied volatility offset or with any of the four offsets.

Table 4
Percentage of months without stochastic dominance violations in the two-period case

	A: 8605–8710	B: 8807–9103	C: 9104–9308	D: 9309–9512	E: 9702–9912	F: 0002–0305	G: 0306–0605
Number of months	15	29	28	26	35	37	36
Unconditional index return distribution, 1928–1986	60 (73)	52 (66)	39 (46)	8 (0)	17 (23)	24 (22)	14 (14)
Unconditional index return distribution, 1972–1986	33 (53)	41 (48)	43 (54)	19 (15)	11 (9)	3 (8)	8 (11)
Unconditional index return distribution, 1987–2006	13 (27)	45 (48)	46 (61)	23 (27)	3 (0)	0 (8)	8 (8)
Unconditional index return distribution, 1988–2006	13 (20)	45 (55)	57 (75)	31 (31)	6 (6)	5 (8)	8 (8)
Conditional index return distribution, 1972–2006, Based on GARCH (1,1)	20 (40)	34 (41)	50 (82)	58 (58)	3 (9)	14 (11)	6 (6)
Conditional index return distribution, 1972–2006, based on implied vol.	87 (93)	66 (66)	61 (75)	54 (58)	14 (14)	27 (30)	8 (14)
Conditional index return distribution, 1972–2006, based on VIX	N/A	N/A	64 (75)	42 (50)	9 (9)	14 (16)	8 (11)

The table displays the percentage of months in which stochastic dominance violations are absent in the cross section of option prices *when one intermediate trading date is allowed* over the life of the 1-month options. The one-way transaction costs rate on the index is 50 bps. The transaction costs on the options are proportional; for the ATM calls they are 10 bps of the index price. In parentheses, the table displays the percentage of months in which stochastic dominance violations are absent in the case when *no intermediate trading is allowed* over the life of the 1-month options. Two periods of 15 days are used.

Read:

- Table 3 deliberately shifts the ATM implied volatility by -2, -1, +1, and +2 percentage points. This is a heavy-handed robustness check designed to see whether a small IV bias drives the violations.
- In panel A, raising IV by 2% can eliminate precrash violations. But the best-of-above row still has low feasibility after 1997: 29% in panel E, 38% in panel F, and 19% in panel G.
- Table 4 allows one intermediate trading date. Numbers outside parentheses are the two-period results; parentheses show the no-intermediate-trading comparison under the same 15-day-return construction.
- In most cases, intermediate trading lowers feasibility and therefore increases the incidence of violations.

Economic message: The results survive generous volatility and trading-opportunity adjustments. More trading flexibility does not rescue the option prices; it often makes the violation easier to exploit.

5.4 Figures 1 and 2: bound violations in the Berkeley-data periods

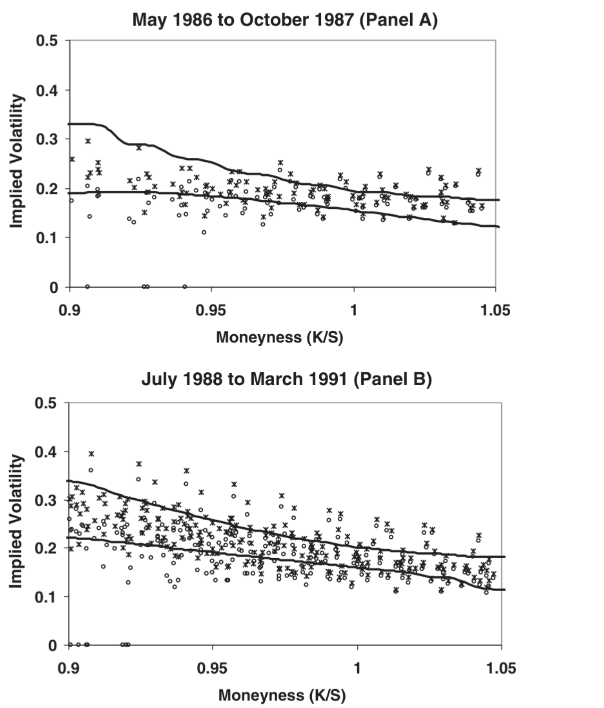

Figure 1
Bound violations over May 1986 to October 1987 and July 1988 to March 1991
The observed bid (circles) and ask (crosses) call prices, as implied volatilities, are plotted as functions of the moneyness. The upper and lower option bounds are based on the index sample distribution for 1972–2006, rescaled with the conditional volatility of the relevant panel. The transaction costs rate on the index is 50 bps.

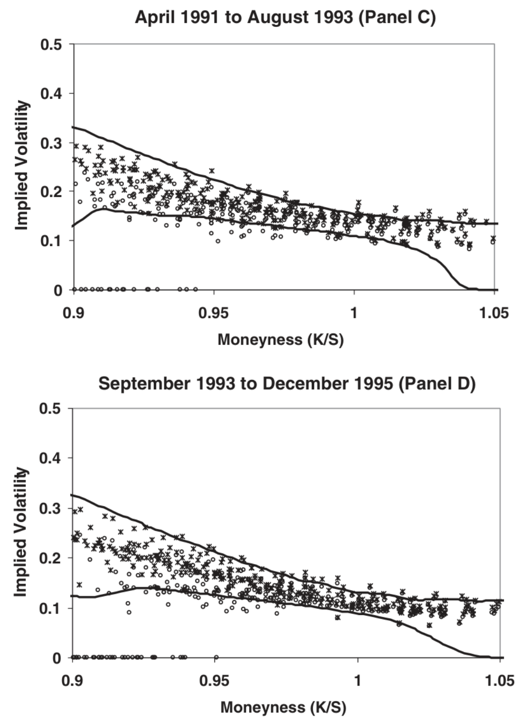

Figure 2
Bound violations over April 1991 to August 1993 and September 1993 to December 1995
The observed bid (circles) and ask (crosses) call prices, as implied volatilities, are plotted as functions of the moneyness. The upper and lower option bounds are based on the index sample distribution for 1972–2006, rescaled with the conditional volatility of the relevant panel. The transaction costs rate on the index is 50 bps.

Read:

- The figures plot bid prices as circles and ask prices as crosses, expressed as implied volatilities.
- The two solid curves are the stochastic-dominance upper and lower bounds.
- In panel A, some OTM bid prices lie above the upper bound and some ITM ask prices lie below the lower bound.
- Panels C and D show fewer violations when the conditional distribution is based on implied volatility, consistent with the table evidence for 1991–1995.

Economic message: Precrash option prices can look close to BSM and still fail a deeper economic test. The implied-volatility surface alone is not enough to determine rationality.

5.5 Figures 3 and 4: bound violations in the OptionMetrics periods

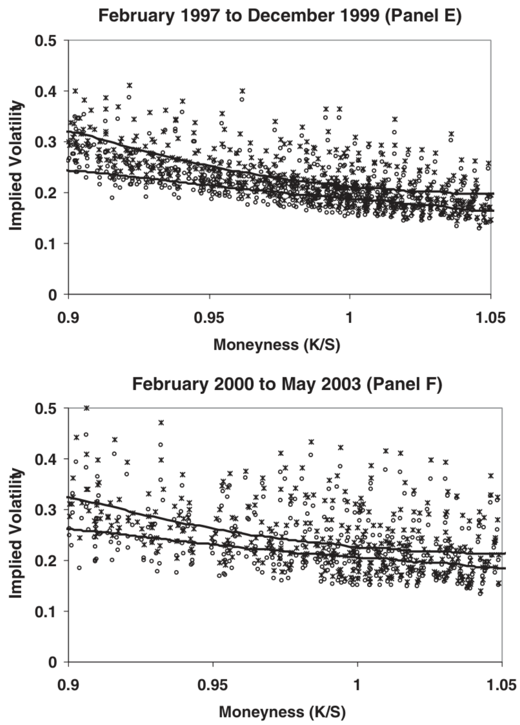

Figure 3
Bound violations over February 1997 to December 1999 and February 2000 to May 2003
The observed bid (circles) and ask (crosses) call prices, as implied volatilities, are plotted as functions of the moneyness. The upper and lower option bounds are based on the index sample distribution for 1972–2006, rescaled with the conditional volatility of the relevant panel. The transaction costs rate on the index is 50 bps.

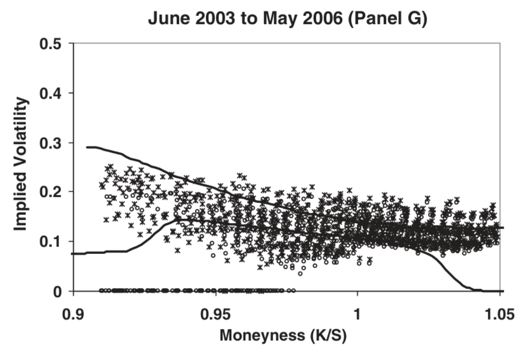

Figure 4
Bound violations over June 2003 to May 2006
The observed bid (circles) and ask (crosses) call prices, as implied volatilities, are plotted as functions of the moneyness. The upper and lower option bounds are based on the index sample distribution for 1972–2006, rescaled with the conditional volatility of the relevant panel. The transaction costs rate on the index is 50 bps.

Read:

- Panels E and F show many bid prices above the upper bound, for both ITM and OTM calls.
- Panel G has little obvious smile, yet bound violations persist.
- The later-period violations are not symmetric around the bounds: they are mostly upper-bound violations rather than an equal mix of upper and lower misses.

Economic message: The evidence does not look like random low-quality data. It looks like systematic overpricing of some calls relative to stochastic-dominance bounds, especially during 1997–2003.

6 Short summary

- The paper evaluates S&P 500 index call prices using stochastic dominance rather than only BSM fit.
- The key restriction is the existence of a positive, decreasing pricing kernel for at least one risk-averse marginal trader with wealth increasing in the market return.
- The test uses a linear program for the full option cross section and separate upper/lower bounds for individual options.
- The data cover 1-month calls from 1986–2006, split into seven subperiods around the 1987 crash and the data-source change.
- Index-return distributions are estimated nonparametrically from several samples and conditionally with GARCH, IV, and VIX.
- Violations appear even when transaction costs are large and when option-implied volatility is used to scale the return distribution.
- Precrash BSM-looking prices are not necessarily rational: the bounds suggest that a smile would actually have reduced some precrash violations.

- Postcrash violations involve both ITM and OTM calls, contradicting a simple left-tail-only interpretation.
- Intermediate trading generally increases the incidence of violations.
- Later-period violations rise again after 1997 and are concentrated above the upper bound, weakening the claim that option markets steadily became more rational after the crash.

7 Conclusion

7.1 Core conclusion

S&P 500 index option prices frequently violate broad stochastic-dominance restrictions. The failures remain after allowing for market incompleteness, bid-ask spreads, transaction costs, nonparametric physical return distributions, conditional volatility, and some state dependence.

7.2 Economic intuition

If a call bid price is too high relative to the upper bound, a trader can write that call and combine it with positions in the index and bond to obtain a payoff profile that improves expected utility. If a call ask price is too low relative to the lower bound, a trader can buy the call and again improve expected utility. The test therefore converts option-price anomalies into utility-improving trading opportunities for a broad class of risk-averse traders.

7.3 Takeaway

The paper's message is stronger than "BSM fails." It says that even after relaxing BSM's most restrictive assumptions, many observed S&P 500 call prices cannot be reconciled with basic monotonicity restrictions implied by risk aversion and market-return-linked wealth.